

```
In [ ]: import pandas as pd

# Load CSV file into a DataFrame
df = pd.read_csv('/content/drive/MyDrive/df_merged_cleaned.csv')

# Display the first 5 rows
print(df.head())
df.shape
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	FLAG_OWN_CAR	FLAG_OWN_REALTY	\
0	100002	1	0	0	1	
1	100003	0	0	0	0	
2	100004	0	1	1	1	
3	100006	0	0	0	1	
4	100007	0	0	0	1	

	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	\
0	0	202500.0	406597.5	24700.5	351000.0	
1	0	270000.0	1293502.5	35698.5	1129500.0	
2	0	67500.0	135000.0	6750.0	135000.0	
3	0	135000.0	312682.5	29686.5	297000.0	
4	0	121500.0	513000.0	21865.5	513000.0	

	SK_DPD_DEF_max	SK_DPD_DEF_min	SK_DPD_DEF_count	\
0	NaN	NaN	NaN	
1	NaN	NaN	NaN	
2	NaN	NaN	NaN	
3	0.0	0.0	6.0	
4	NaN	NaN	NaN	

	NAME_CONTRACT_STATUS_Active_y	NAME_CONTRACT_STATUS_Approved	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	1.0	0.0	
4	NaN	NaN	

	NAME_CONTRACT_STATUS_Completed_y	NAME_CONTRACT_STATUS_Demand_y	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	0.0	0.0	
4	NaN	NaN	

	NAME_CONTRACT_STATUS_Refused_y	NAME_CONTRACT_STATUS_Sent proposal	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	0.0	0.0	
4	NaN	NaN	

	NAME_CONTRACT_STATUS_Signed_y
0	NaN
1	NaN
2	NaN
3	0.0
4	NaN

[5 rows x 657 columns]

```
Out[ ]: (307511, 657)
```

```
In [ ]: # Check data types
print(df.dtypes)

# Check for missing values
print(df.isnull().sum())
```

```

SK_ID_CURR          int64
TARGET              int64
NAME_CONTRACT_TYPE  int64
FLAG_OWN_CAR        int64
FLAG_OWN_REALTY     int64
...
NAME_CONTRACT_STATUS_Completed_y  float64
NAME_CONTRACT_STATUS_Demand_y     float64
NAME_CONTRACT_STATUS_Refused_y     float64
NAME_CONTRACT_STATUS_Sent proposal float64
NAME_CONTRACT_STATUS_Signed_y      float64
Length: 657, dtype: object
SK_ID_CURR          0
TARGET              0
NAME_CONTRACT_TYPE  0
FLAG_OWN_CAR        0
FLAG_OWN_REALTY     0
...
NAME_CONTRACT_STATUS_Completed_y  220606
NAME_CONTRACT_STATUS_Demand_y     220606
NAME_CONTRACT_STATUS_Refused_y     220606
NAME_CONTRACT_STATUS_Sent proposal 220606
NAME_CONTRACT_STATUS_Signed_y      220606
Length: 657, dtype: int64

```

```

In [ ]: # Check for new missing values after the join
        print("Missing values after join:")
        print(df.isnull().sum().sort_values(ascending=False))

```

```

Missing values after join:
SK_DPD_sum          220606
NAME_CONTRACT_STATUS_Signed_y  220606
SK_DPD_mean         220606
CNT_INSTALMENT_MATURE_CUM_count  220606
CNT_INSTALMENT_MATURE_CUM_min  220606
...
LIVE_REGION_NOT_WORK_REGION    0
REG_CITY_NOT_LIVE_CITY         0
REG_CITY_NOT_WORK_CITY         0
LIVE_CITY_NOT_WORK_CITY        0
EXT_SOURCE_1                   0
Length: 657, dtype: int64

```

```

In [ ]: categorical_cols = df.select_dtypes(include=['object']).columns.tolist()
        numeric_cols = df.select_dtypes(include=['number']).columns.tolist()

```

```

In [ ]: print(f"Categorical columns: {categorical_cols}")
        print(f"Numeric columns: {len(numeric_cols)} columns")

```

```

Categorical columns: []
Numeric columns: 537 columns

```

```

In [ ]: print("Missing values check:")
        print(df.isnull().sum().sum())

```

```

Missing values check:
30864083

```

```

In [ ]: print("Data types:")
        print(df.dtypes.value_counts())

```

```

Data types:
float64    491
bool       120
int64      46
Name: count, dtype: int64

```

```

In [ ]: # Check missing values per column
        missing_per_column = df.isnull().sum().sort_values(ascending=False)
        missing_percentage = (missing_per_column / len(df)) * 100

        print("Top 20 columns with most missing values:")
        for col in missing_per_column.head(20).index:
            count = missing_per_column[col]
            percent = missing_percentage[col]
            print(f"{col}: {count:,} missing ({percent:.1f}%)")

```

Top 20 columns with most missing values:

- SK_DPD_sum: 220,606 missing (71.7%)
- NAME_CONTRACT_STATUS_Signed_y: 220,606 missing (71.7%)
- SK_DPD_mean: 220,606 missing (71.7%)
- CNT_INSTALMENT_MATURE_CUM_count: 220,606 missing (71.7%)
- CNT_INSTALMENT_MATURE_CUM_min: 220,606 missing (71.7%)
- CNT_INSTALMENT_MATURE_CUM_max: 220,606 missing (71.7%)
- CNT_INSTALMENT_MATURE_CUM_sum: 220,606 missing (71.7%)
- CNT_INSTALMENT_MATURE_CUM_mean: 220,606 missing (71.7%)
- CNT_DRAWINGS_POS_CURRENT_count: 220,606 missing (71.7%)
- CNT_DRAWINGS_POS_CURRENT_min: 220,606 missing (71.7%)
- AMT_RECEIVABLE_PRINCIPAL_mean: 220,606 missing (71.7%)
- AMT_RECEIVABLE_PRINCIPAL_count: 220,606 missing (71.7%)
- AMT_RECEIVABLE_PRINCIPAL_sum: 220,606 missing (71.7%)
- AMT_RECEIVABLE_PRINCIPAL_max: 220,606 missing (71.7%)
- AMT_RECEIVABLE_PRINCIPAL_min: 220,606 missing (71.7%)
- AMT_RECIVABLE_sum: 220,606 missing (71.7%)
- AMT_RECIVABLE_mean: 220,606 missing (71.7%)
- AMT_RECIVABLE_max: 220,606 missing (71.7%)
- AMT_RECIVABLE_min: 220,606 missing (71.7%)
- AMT_DRAWINGS_OTHER_CURRENT_sum: 220,606 missing (71.7%)

```
In [ ]: # Drop columns with more than 70% missing data
threshold = 0.7
cols_to_drop = missing_percentage[missing_percentage > threshold * 100].index
print(f"Columns to drop (>{threshold*100}% missing): {len(cols_to_drop)}")
df_cleaned = df.drop(columns=cols_to_drop)
print(f"Shape after dropping high-missing columns: {df_cleaned.shape}")
```

Columns to drop (>70.0% missing): 107
Shape after dropping high-missing columns: (307511, 550)

```
In [ ]: # For boolean columns
for col in df_cleaned.select_dtypes(include='bool').columns:
    if df_cleaned[col].isnull().sum() > 0:
        # Use mode
        mode_val = df_cleaned[col].mode().iloc[0] if not df_cleaned[col].mode().empty else False
        df_cleaned[col] = df_cleaned[col].fillna(mode_val)

# For numeric columns - you can choose between mean/median
for col in df_cleaned.select_dtypes(include=['int64', 'float64']).columns:
    if df_cleaned[col].isnull().sum() > 0:
        # Use median for robustness against outliers
        median_val = df_cleaned[col].median()
        df_cleaned[col] = df_cleaned[col].fillna(median_val)
```

```
In [ ]: print("Missing values after imputation:", df_cleaned.isnull().sum().sum())
```

Missing values after imputation: 0

```
In [ ]: # Final check
print("Final dataset shape:", df_cleaned.shape)
print("Final missing values:", df_cleaned.isnull().sum().sum())
print("Data types:", df_cleaned.dtypes.value_counts())
```

Final dataset shape: (307511, 550)
Final missing values: 0
Data types: float64 384
bool 120
int64 46
Name: count, dtype: int64

```
In [ ]: # Convert boolean to int (0/1) for better compatibility
bool_cols = df_cleaned.select_dtypes(include='bool').columns
df_cleaned[bool_cols] = df_cleaned[bool_cols].astype(int)

print("Final data types after boolean conversion:")
print(df_cleaned.dtypes.value_counts())
```

Final data types after boolean conversion:
float64 384
int64 166
Name: count, dtype: int64

```
In [ ]: # Save your cleaned dataset to CSV
df.to_csv('/content/drive/MyDrive/home-credit-default-risk/cleaned_dataset.csv', index=False)
print("Dataset saved successfully!")
print(f"Saved dataset shape: {df_cleaned.shape}")
```

Dataset saved successfully!
Saved dataset shape: (307511, 550)

In []:

After this we used this dataset to create df_final_strategic.csv in df_final_strategic.ipynb

In []:

```
import pandas as pd

# Load CSV file into a DataFrame
df_cleaned = pd.read_csv('/content/drive/MyDrive/home-credit-default-risk/df_final_strategic.csv') # Replace

# Display the first 5 rows
print(df_cleaned.head())
df_cleaned.shape
```

	SK_ID_CURR	TARGET	NAME_CONTRACT_TYPE	FLAG_OWN_CAR	FLAG_OWN_REALTY	\
0	100002	1	0	0	1	
1	100003	0	0	0	0	
2	100004	0	1	1	1	
3	100006	0	0	0	1	
4	100007	0	0	0	1	

	CNT_CHILDREN	AMT_INCOME_TOTAL	AMT_CREDIT	AMT_ANNUITY	AMT_GOODS_PRICE	\
0	0	202500.0	406597.5	24700.5	351000.0	
1	0	270000.0	1293502.5	35698.5	1129500.0	
2	0	67500.0	135000.0	6750.0	135000.0	
3	0	135000.0	312682.5	29686.5	297000.0	
4	0	121500.0	513000.0	21865.5	513000.0	

	...	POS_SK_DPD_DEF_SUM	NAME_CONTRACT_STATUS_Active_x	\
0	...	0.0	1.000000	
1	...	0.0	0.928571	
2	...	0.0	0.750000	
3	...	0.0	0.857143	
4	...	0.0	0.939394	

	NAME_CONTRACT_STATUS_Amortized_debt	NAME_CONTRACT_STATUS_Approved_y	\
0	0.0	0.0	
1	0.0	0.0	
2	0.0	0.0	
3	0.0	0.0	
4	0.0	0.0	

	NAME_CONTRACT_STATUS_Canceled_y	NAME_CONTRACT_STATUS_Completed_x	\
0	0.0	0.000000	
1	0.0	0.071429	
2	0.0	0.250000	
3	0.0	0.095238	
4	0.0	0.045455	

	NAME_CONTRACT_STATUS_Demand_x	NAME_CONTRACT_STATUS_Returned to the store	\
0	0.0	0.000000	
1	0.0	0.000000	
2	0.0	0.000000	
3	0.0	0.047619	
4	0.0	0.000000	

	NAME_CONTRACT_STATUS_Signed_x	NAME_CONTRACT_STATUS_XNA
0	0.000000	0.0
1	0.000000	0.0
2	0.000000	0.0
3	0.000000	0.0
4	0.015152	0.0

[5 rows x 364 columns]

Out[]: (307511, 364)

In []:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split

X = df_cleaned.drop('TARGET', axis=1)
y = df_cleaned['TARGET']
```

```

print(f"Features shape: {X.shape}")
print(f"Target shape: {y.shape}")

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

rf = RandomForestClassifier(
    n_estimators=100,
    random_state=42,
    n_jobs=-1
)

print("Training Random Forest for feature importance...")
rf.fit(X_train, y_train)

feature_importance = pd.DataFrame({
    'feature': X.columns,
    'importance': rf.feature_importances_
}).sort_values('importance', ascending=False)

print("Training completed!")

```

```

Features shape: (307511, 363)
Target shape: (307511,)
Training Random Forest for feature importance...
Training completed!

```

```

In [ ]: plt.figure(figsize=(12, 8))
top_features = feature_importance.head(20)

sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
plt.title('Top 20 Most Important Features (Random Forest)', fontsize=16, fontweight='bold')
plt.xlabel('Feature Importance', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.tight_layout()
plt.show()

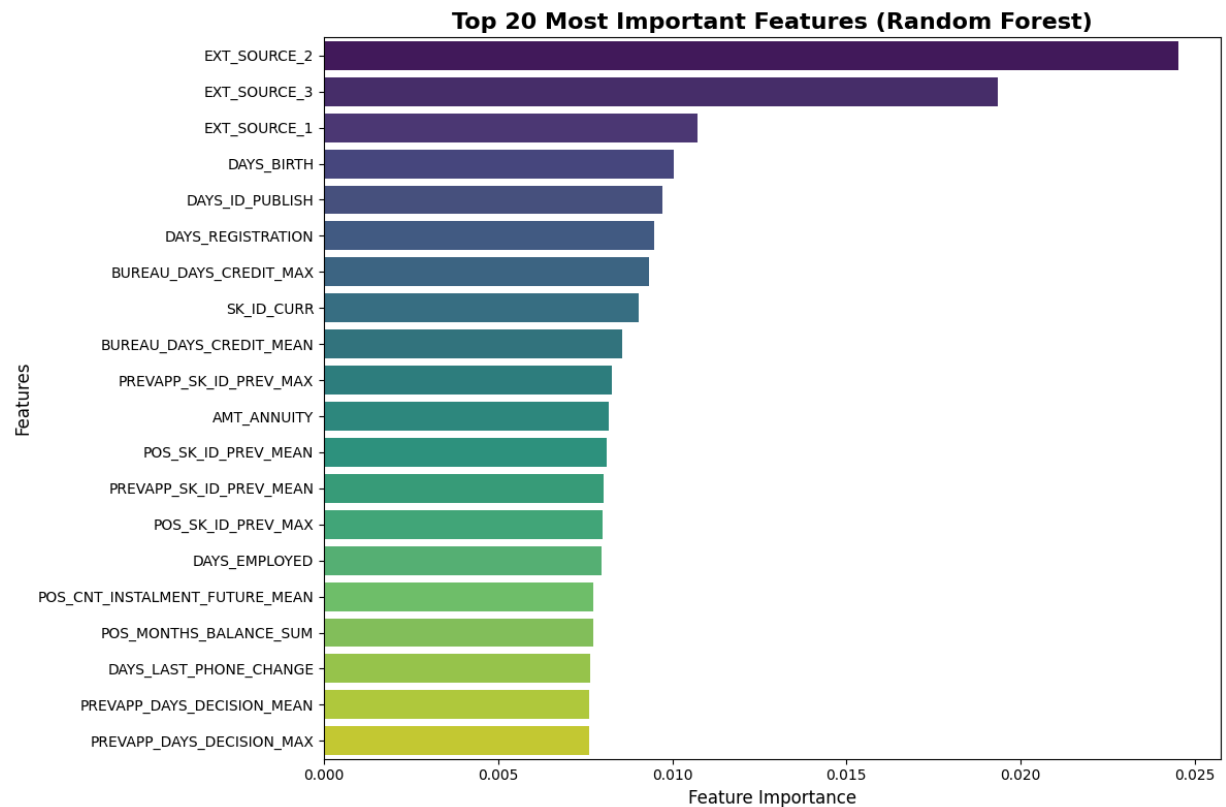
print("\nTop 20 Most Important Features:")
print("="*50)
for i, (idx, row) in enumerate(top_features.iterrows(), 1):
    print(f"{i:2d}. {row['feature']:<40} {row['importance']:.4f}")

```

<ipython-input-3-1630655910>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
```



Top 20 Most Important Features:

```
=====
1. EXT_SOURCE_2                0.0245
2. EXT_SOURCE_3                0.0193
3. EXT_SOURCE_1                0.0107
4. DAYS_BIRTH                  0.0100
5. DAYS_ID_PUBLISH             0.0097
6. DAYS_REGISTRATION           0.0095
7. BUREAU_DAYS_CREDIT_MAX      0.0093
8. SK_ID_CURR                  0.0090
9. BUREAU_DAYS_CREDIT_MEAN     0.0086
10. PREVAPP_SK_ID_PREV_MAX     0.0083
11. AMT_ANNUITY                0.0082
12. POS_SK_ID_PREV_MEAN        0.0081
13. PREVAPP_SK_ID_PREV_MEAN    0.0080
14. POS_SK_ID_PREV_MAX         0.0080
15. DAYS_EMPLOYED              0.0080
16. POS_CNT_INSTALMENT_FUTURE_MEAN 0.0077
17. POS_MONTHS_BALANCE_SUM     0.0077
18. DAYS_LAST_PHONE_CHANGE     0.0076
19. PREVAPP_DAYS_DECISION_MEAN 0.0076
20. PREVAPP_DAYS_DECISION_MAX  0.0076
```

```
In [ ]: plt.figure(figsize=(12, 8))
top_features = feature_importance.head(30)

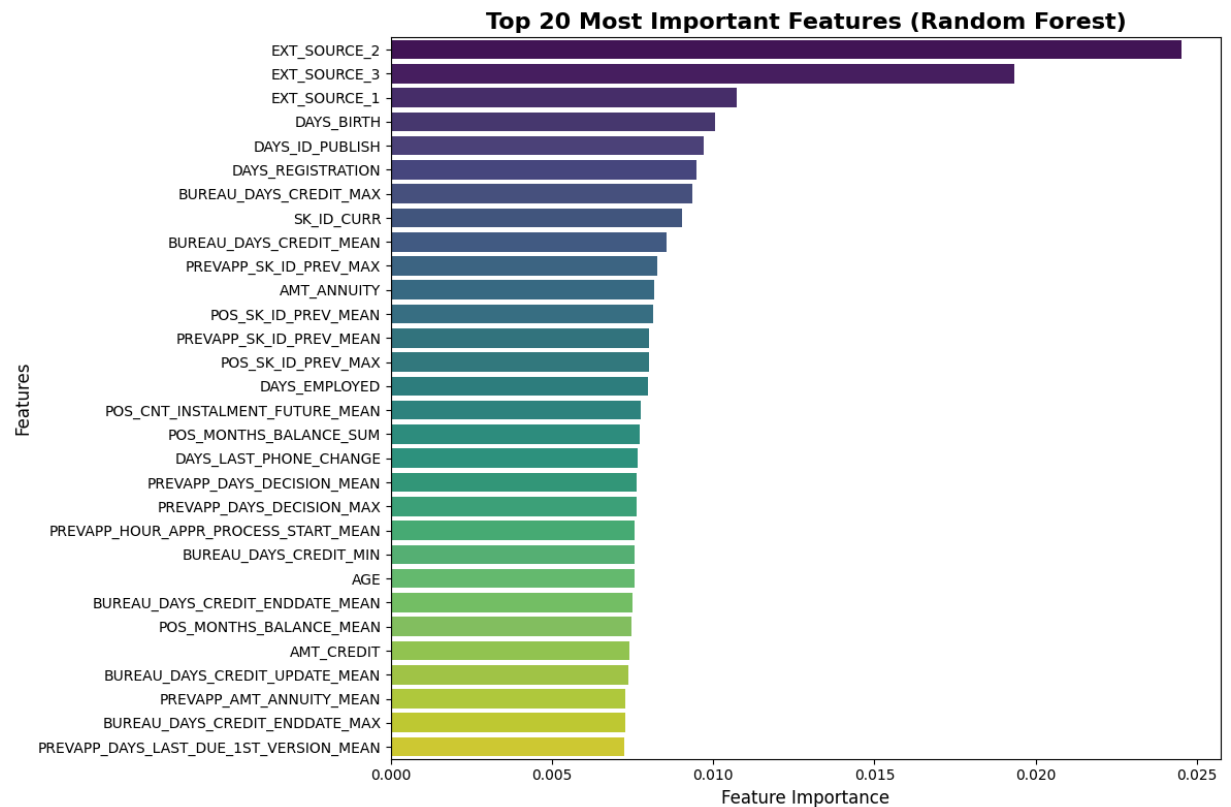
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
plt.title('Top 20 Most Important Features (Random Forest)', fontsize=16, fontweight='bold')
plt.xlabel('Feature Importance', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.tight_layout()
plt.show()

print("\nTop 30 Most Important Features:")
print("="*50)
for i, (idx, row) in enumerate(top_features.iterrows(), 1):
    print(f"{i:2d}. {row['feature']:<40} {row['importance']:.4f}")
```

<ipython-input-4-4153336299>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
```



Top 30 Most Important Features:

=====	
1. EXT_SOURCE_2	0.0245
2. EXT_SOURCE_3	0.0193
3. EXT_SOURCE_1	0.0107
4. DAYS_BIRTH	0.0100
5. DAYS_ID_PUBLISH	0.0097
6. DAYS_REGISTRATION	0.0095
7. BUREAU_DAYS_CREDIT_MAX	0.0093
8. SK_ID_CURR	0.0090
9. BUREAU_DAYS_CREDIT_MEAN	0.0086
10. PREVAPP_SK_ID_PREV_MAX	0.0083
11. AMT_ANNUITY	0.0082
12. POS_SK_ID_PREV_MEAN	0.0081
13. PREVAPP_SK_ID_PREV_MEAN	0.0080
14. POS_SK_ID_PREV_MAX	0.0080
15. DAYS_EMPLOYED	0.0080
16. POS_CNT_INSTALMENT_FUTURE_MEAN	0.0077
17. POS_MONTHS_BALANCE_SUM	0.0077
18. DAYS_LAST_PHONE_CHANGE	0.0076
19. PREVAPP_DAYS_DECISION_MEAN	0.0076
20. PREVAPP_DAYS_DECISION_MAX	0.0076
21. PREVAPP_HOUR_APPR_PROCESS_START_MEAN	0.0076
22. BUREAU_DAYS_CREDIT_MIN	0.0075
23. AGE	0.0075
24. BUREAU_DAYS_CREDIT_ENDDATE_MEAN	0.0075
25. POS_MONTHS_BALANCE_MEAN	0.0074
26. AMT_CREDIT	0.0074
27. BUREAU_DAYS_CREDIT_UPDATE_MEAN	0.0074
28. PREVAPP_AMT_ANNUITY_MEAN	0.0073
29. BUREAU_DAYS_CREDIT_ENDDATE_MAX	0.0073
30. PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MEAN	0.0072

```
In [ ]: plt.figure(figsize=(12, 8))
top_features = feature_importance.head(40)

sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
plt.title('Top 20 Most Important Features (Random Forest)', fontsize=16, fontweight='bold')
plt.xlabel('Feature Importance', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.tight_layout()
plt.show()

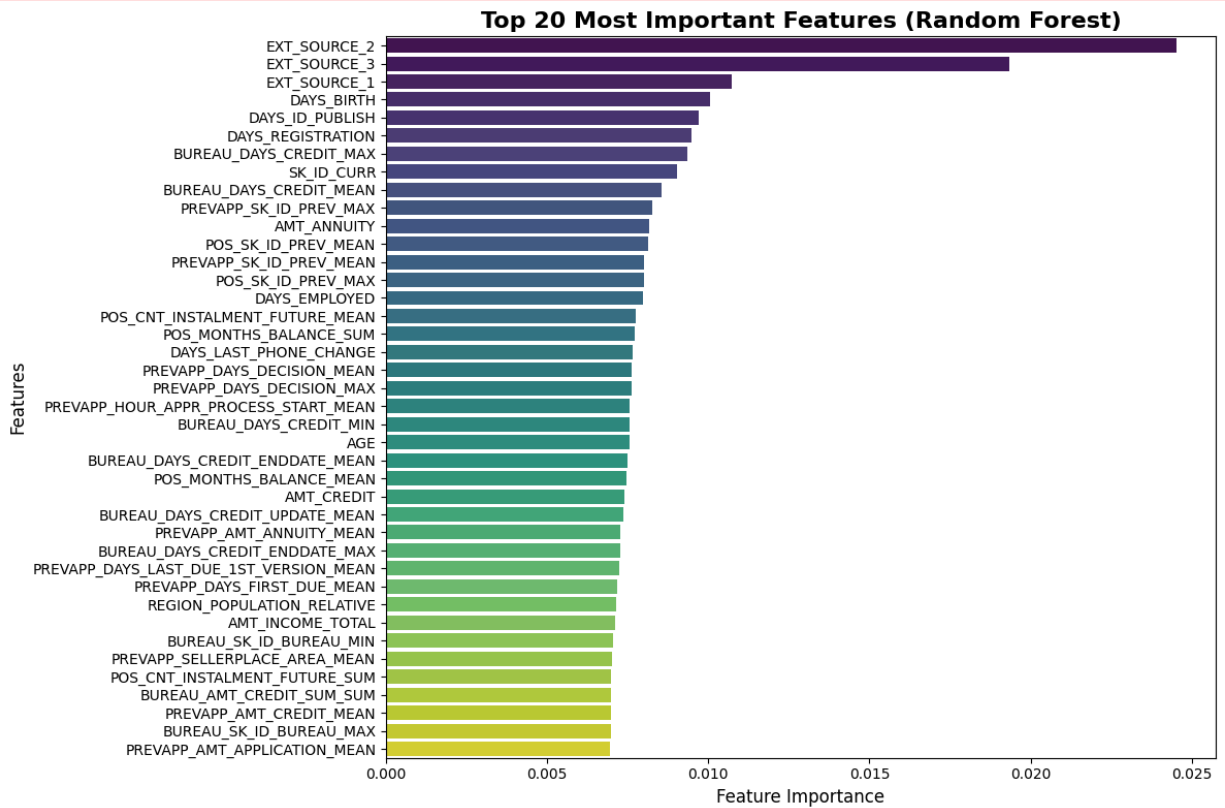
print("\nTop 40 Most Important Features:")
print("="*50)
```

```
for i, (idx, row) in enumerate(top_features.iterrows(), 1):
    print(f"{i:2d}. {row['feature']:<40} {row['importance']:.4f}")
```

<ipython-input-5-726849988>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
```



Top 40 Most Important Features:

=====		
1.	EXT_SOURCE_2	0.0245
2.	EXT_SOURCE_3	0.0193
3.	EXT_SOURCE_1	0.0107
4.	DAYS_BIRTH	0.0100
5.	DAYS_ID_PUBLISH	0.0097
6.	DAYS_REGISTRATION	0.0095
7.	BUREAU_DAYS_CREDIT_MAX	0.0093
8.	SK_ID_CURR	0.0090
9.	BUREAU_DAYS_CREDIT_MEAN	0.0086
10.	PREVAPP_SK_ID_PREV_MAX	0.0083
11.	AMT_ANNUITY	0.0082
12.	POS_SK_ID_PREV_MEAN	0.0081
13.	PREVAPP_SK_ID_PREV_MEAN	0.0080
14.	POS_SK_ID_PREV_MAX	0.0080
15.	DAYS_EMPLOYED	0.0080
16.	POS_CNT_INSTALMENT_FUTURE_MEAN	0.0077
17.	POS_MONTHS_BALANCE_SUM	0.0077
18.	DAYS_LAST_PHONE_CHANGE	0.0076
19.	PREVAPP_DAYS_DECISION_MEAN	0.0076
20.	PREVAPP_DAYS_DECISION_MAX	0.0076
21.	PREVAPP_HOUR_APPR_PROCESS_START_MEAN	0.0076
22.	BUREAU_DAYS_CREDIT_MIN	0.0075
23.	AGE	0.0075
24.	BUREAU_DAYS_CREDIT_ENDDATE_MEAN	0.0075
25.	POS_MONTHS_BALANCE_MEAN	0.0074
26.	AMT_CREDIT	0.0074
27.	BUREAU_DAYS_CREDIT_UPDATE_MEAN	0.0074
28.	PREVAPP_AMT_ANNUITY_MEAN	0.0073
29.	BUREAU_DAYS_CREDIT_ENDDATE_MAX	0.0073
30.	PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MEAN	0.0072
31.	PREVAPP_DAYS_FIRST_DUE_MEAN	0.0072
32.	REGION_POPULATION_RELATIVE	0.0072
33.	AMT_INCOME_TOTAL	0.0071
34.	BUREAU_SK_ID_BUREAU_MIN	0.0070
35.	PREVAPP_SELLERPLACE_AREA_MEAN	0.0070
36.	POS_CNT_INSTALMENT_FUTURE_SUM	0.0070
37.	BUREAU_AMT_CREDIT_SUM_SUM	0.0070
38.	PREVAPP_AMT_CREDIT_MEAN	0.0070
39.	BUREAU_SK_ID_BUREAU_MAX	0.0070
40.	PREVAPP_AMT_APPLICATION_MEAN	0.0070

```
In [ ]: plt.figure(figsize=(12, 8))
top_features = feature_importance.head(100)

sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
plt.title('Top 20 Most Important Features (Random Forest)', fontsize=16, fontweight='bold')
plt.xlabel('Feature Importance', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.tight_layout()
plt.show()

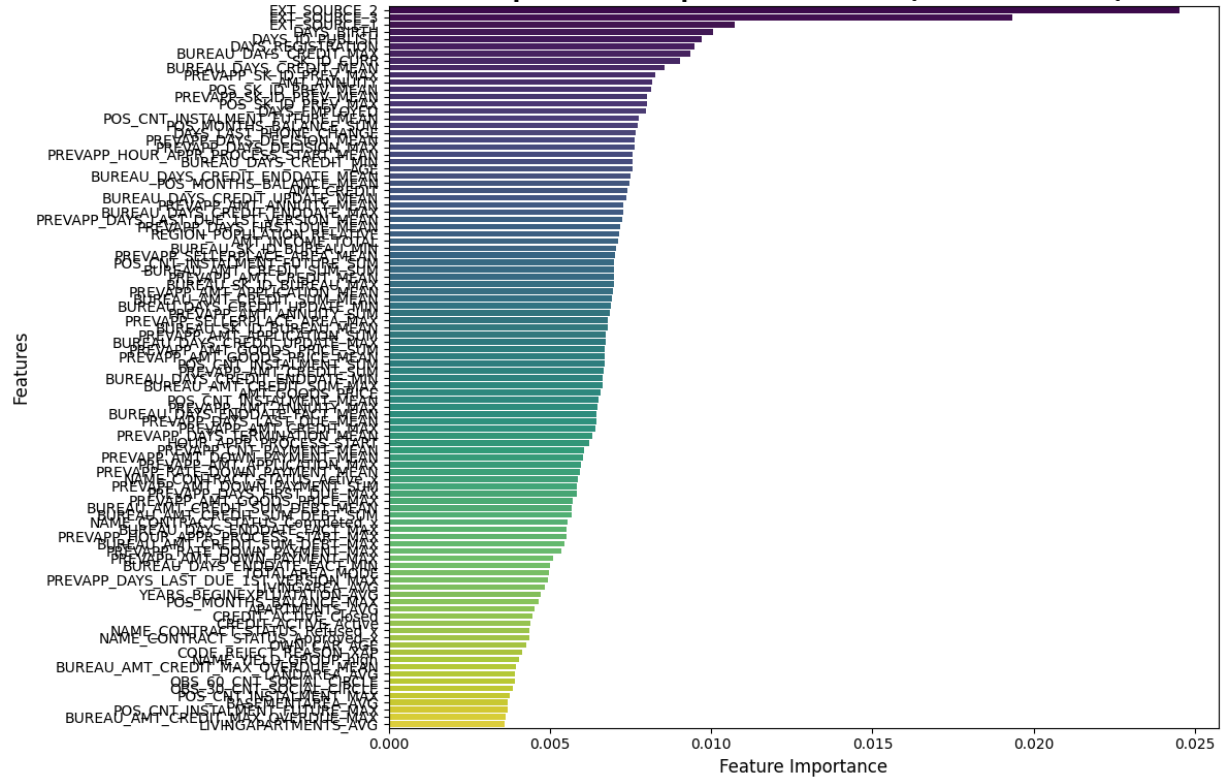
print("\nTop 20 Most Important Features:")
print("="*50)
for i, (idx, row) in enumerate(top_features.iterrows(), 1):
    print(f"{i:2d}. {row['feature']:<40} {row['importance']:.4f}")
```

<ipython-input-6-3814755389>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
```

Top 20 Most Important Features (Random Forest)



Top 20 Most Important Features:

=====	
1. EXT_SOURCE_2	0.0245
2. EXT_SOURCE_3	0.0193
3. EXT_SOURCE_1	0.0107
4. DAYS_BIRTH	0.0100
5. DAYS_ID_PUBLISH	0.0097
6. DAYS_REGISTRATION	0.0095
7. BUREAU_DAYS_CREDIT_MAX	0.0093
8. SK_ID_CURR	0.0090
9. BUREAU_DAYS_CREDIT_MEAN	0.0086
10. PREVAPP_SK_ID_PREV_MAX	0.0083
11. AMT_ANNUITY	0.0082
12. POS_SK_ID_PREV_MEAN	0.0081
13. PREVAPP_SK_ID_PREV_MEAN	0.0080
14. POS_SK_ID_PREV_MAX	0.0080
15. DAYS_EMPLOYED	0.0080
16. POS_CNT_INSTALMENT_FUTURE_MEAN	0.0077
17. POS_MONTHS_BALANCE_SUM	0.0077
18. DAYS_LAST_PHONE_CHANGE	0.0076
19. PREVAPP_DAYS_DECISION_MEAN	0.0076
20. PREVAPP_DAYS_DECISION_MAX	0.0076
21. PREVAPP_HOUR_APPR_PROCESS_START_MEAN	0.0076
22. BUREAU_DAYS_CREDIT_MIN	0.0075
23. AGE	0.0075
24. BUREAU_DAYS_CREDIT_ENDDATE_MEAN	0.0075
25. POS_MONTHS_BALANCE_MEAN	0.0074
26. AMT_CREDIT	0.0074
27. BUREAU_DAYS_CREDIT_UPDATE_MEAN	0.0074
28. PREVAPP_AMT_ANNUITY_MEAN	0.0073
29. BUREAU_DAYS_CREDIT_ENDDATE_MAX	0.0073
30. PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MEAN	0.0072
31. PREVAPP_DAYS_FIRST_DUE_MEAN	0.0072
32. REGION_POPULATION_RELATIVE	0.0072
33. AMT_INCOME_TOTAL	0.0071
34. BUREAU_SK_ID_BUREAU_MIN	0.0070
35. PREVAPP_SELLERPLACE_AREA_MEAN	0.0070
36. POS_CNT_INSTALMENT_FUTURE_SUM	0.0070
37. BUREAU_AMT_CREDIT_SUM_SUM	0.0070
38. PREVAPP_AMT_CREDIT_MEAN	0.0070
39. BUREAU_SK_ID_BUREAU_MAX	0.0070
40. PREVAPP_AMT_APPLICATION_MEAN	0.0070
41. BUREAU_AMT_CREDIT_SUM_MEAN	0.0069
42. BUREAU_DAYS_CREDIT_UPDATE_MIN	0.0069
43. PREVAPP_AMT_ANNUITY_SUM	0.0068
44. PREVAPP_SELLERPLACE_AREA_MAX	0.0068
45. BUREAU_SK_ID_BUREAU_MEAN	0.0068
46. PREVAPP_AMT_APPLICATION_SUM	0.0067
47. BUREAU_DAYS_CREDIT_UPDATE_MAX	0.0067
48. PREVAPP_AMT_GOODS_PRICE_SUM	0.0067
49. PREVAPP_AMT_GOODS_PRICE_MEAN	0.0067
50. POS_CNT_INSTALMENT_SUM	0.0067
51. PREVAPP_AMT_CREDIT_SUM	0.0066
52. BUREAU_DAYS_CREDIT_ENDDATE_MIN	0.0066
53. BUREAU_AMT_CREDIT_SUM_MAX	0.0066
54. AMT_GOODS_PRICE	0.0066
55. POS_CNT_INSTALMENT_MEAN	0.0065
56. PREVAPP_AMT_ANNUITY_MAX	0.0065
57. BUREAU_DAYS_ENDDATE_FACT_MEAN	0.0064
58. PREVAPP_DAYS_LAST_DUE_MEAN	0.0064
59. PREVAPP_AMT_CREDIT_MAX	0.0064
60. PREVAPP_DAYS_TERMINATION_MEAN	0.0063
61. HOUR_APPR_PROCESS_START	0.0062
62. PREVAPP_CNT_PAYMENT_MEAN	0.0061
63. PREVAPP_AMT_DOWN_PAYMENT_MEAN	0.0060
64. PREVAPP_AMT_APPLICATION_MAX	0.0060
65. PREVAPP_RATE_DOWN_PAYMENT_MEAN	0.0059
66. NAME_CONTRACT_STATUS_Active_x	0.0059
67. PREVAPP_AMT_DOWN_PAYMENT_SUM	0.0058
68. PREVAPP_DAYS_FIRST_DUE_MAX	0.0058
69. PREVAPP_AMT_GOODS_PRICE_MAX	0.0057
70. BUREAU_AMT_CREDIT_SUM_DEBT_MEAN	0.0057
71. BUREAU_AMT_CREDIT_SUM_DEBT_SUM	0.0057
72. NAME_CONTRACT_STATUS_Completed_x	0.0055
73. BUREAU_DAYS_ENDDATE_FACT_MAX	0.0055
74. PREVAPP_HOUR_APPR_PROCESS_START_MAX	0.0055
75. BUREAU_AMT_CREDIT_SUM_DEBT_MAX	0.0054
76. PREVAPP_RATE_DOWN_PAYMENT_MAX	0.0053

77. PREVAPP_AMT_DOWN_PAYMENT_MAX	0.0051
78. BUREAU_DAYS_ENDDATE_FACT_MIN	0.0050
79. TOTALAREA_MODE	0.0050
80. PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MAX	0.0049
81. LIVINGAREA_AVG	0.0048
82. YEARS_BEGINEXPLUATATION_AVG	0.0047
83. POS_MONTHS_BALANCE_MAX	0.0046
84. APARTMENTS_AVG	0.0045
85. CREDIT_ACTIVE_Closed	0.0045
86. CREDIT_ACTIVE_Active	0.0044
87. NAME_CONTRACT_STATUS_Refused_x	0.0044
88. NAME_CONTRACT_STATUS_Approved_x	0.0043
89. OWN_CAR_AGE	0.0043
90. CODE_REJECT_REASON_XAP	0.0041
91. NAME_YIELD_GROUP_high	0.0040
92. BUREAU_AMT_CREDIT_MAX_OVERDUE_MEAN	0.0039
93. LANDAREA_AVG	0.0039
94. OBS_60_CNT_SOCIAL_CIRCLE	0.0039
95. OBS_30_CNT_SOCIAL_CIRCLE	0.0039
96. POS_CNT_INSTALMENT_MAX	0.0037
97. BASEMENTAREA_AVG	0.0037
98. POS_CNT_INSTALMENT_FUTURE_MAX	0.0037
99. BUREAU_AMT_CREDIT_MAX_OVERDUE_MAX	0.0036
100. LIVINGAPARTMENTS_AVG	0.0036

```
In [ ]: plt.figure(figsize=(12, 8))
top_features = feature_importance.head(200)

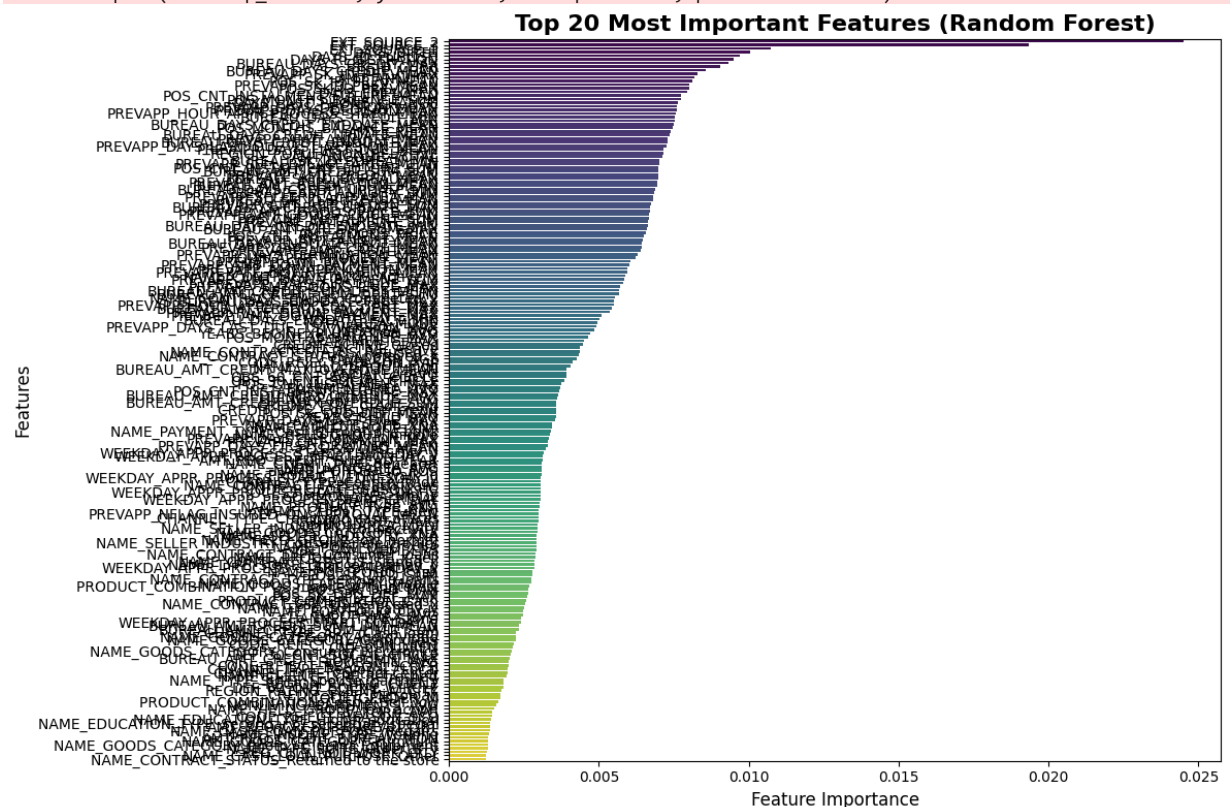
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
plt.title('Top 20 Most Important Features (Random Forest)', fontsize=16, fontweight='bold')
plt.xlabel('Feature Importance', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.tight_layout()
plt.show()

print("\nTop 200 Most Important Features:")
print("="*50)
for i, (idx, row) in enumerate(top_features.iterrows(), 1):
    print(f"{i:2d}. {row['feature']:<40} {row['importance']:.4f}")
```

<ipython-input-7-3996148651>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
```



Top 200 Most Important Features:

=====		
1.	EXT_SOURCE_2	0.0245
2.	EXT_SOURCE_3	0.0193
3.	EXT_SOURCE_1	0.0107
4.	DAYS_BIRTH	0.0100
5.	DAYS_ID_PUBLISH	0.0097
6.	DAYS_REGISTRATION	0.0095
7.	BUREAU_DAYS_CREDIT_MAX	0.0093
8.	SK_ID_CURR	0.0090
9.	BUREAU_DAYS_CREDIT_MEAN	0.0086
10.	PREVAPP_SK_ID_PREV_MAX	0.0083
11.	AMT_ANNUITY	0.0082
12.	POS_SK_ID_PREV_MEAN	0.0081
13.	PREVAPP_SK_ID_PREV_MEAN	0.0080
14.	POS_SK_ID_PREV_MAX	0.0080
15.	DAYS_EMPLOYED	0.0080
16.	POS_CNT_INSTALMENT_FUTURE_MEAN	0.0077
17.	POS_MONTHS_BALANCE_SUM	0.0077
18.	DAYS_LAST_PHONE_CHANGE	0.0076
19.	PREVAPP_DAYS_DECISION_MEAN	0.0076
20.	PREVAPP_DAYS_DECISION_MAX	0.0076
21.	PREVAPP_HOUR_APPR_PROCESS_START_MEAN	0.0076
22.	BUREAU_DAYS_CREDIT_MIN	0.0075
23.	AGE	0.0075
24.	BUREAU_DAYS_CREDIT_ENDDATE_MEAN	0.0075
25.	POS_MONTHS_BALANCE_MEAN	0.0074
26.	AMT_CREDIT	0.0074
27.	BUREAU_DAYS_CREDIT_UPDATE_MEAN	0.0074
28.	PREVAPP_AMT_ANNUITY_MEAN	0.0073
29.	BUREAU_DAYS_CREDIT_ENDDATE_MAX	0.0073
30.	PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MEAN	0.0072
31.	PREVAPP_DAYS_FIRST_DUE_MEAN	0.0072
32.	REGION_POPULATION_RELATIVE	0.0072
33.	AMT_INCOME_TOTAL	0.0071
34.	BUREAU_SK_ID_BUREAU_MIN	0.0070
35.	PREVAPP_SELLERPLACE_AREA_MEAN	0.0070
36.	POS_CNT_INSTALMENT_FUTURE_SUM	0.0070
37.	BUREAU_AMT_CREDIT_SUM_SUM	0.0070
38.	PREVAPP_AMT_CREDIT_MEAN	0.0070
39.	BUREAU_SK_ID_BUREAU_MAX	0.0070
40.	PREVAPP_AMT_APPLICATION_MEAN	0.0070
41.	BUREAU_AMT_CREDIT_SUM_MEAN	0.0069
42.	BUREAU_DAYS_CREDIT_UPDATE_MIN	0.0069
43.	PREVAPP_AMT_ANNUITY_SUM	0.0068
44.	PREVAPP_SELLERPLACE_AREA_MAX	0.0068
45.	BUREAU_SK_ID_BUREAU_MEAN	0.0068
46.	PREVAPP_AMT_APPLICATION_SUM	0.0067
47.	BUREAU_DAYS_CREDIT_UPDATE_MAX	0.0067
48.	PREVAPP_AMT_GOODS_PRICE_SUM	0.0067
49.	PREVAPP_AMT_GOODS_PRICE_MEAN	0.0067
50.	POS_CNT_INSTALMENT_SUM	0.0067
51.	PREVAPP_AMT_CREDIT_SUM	0.0066
52.	BUREAU_DAYS_CREDIT_ENDDATE_MIN	0.0066
53.	BUREAU_AMT_CREDIT_SUM_MAX	0.0066
54.	AMT_GOODS_PRICE	0.0066
55.	POS_CNT_INSTALMENT_MEAN	0.0065
56.	PREVAPP_AMT_ANNUITY_MAX	0.0065
57.	BUREAU_DAYS_ENDDATE_FACT_MEAN	0.0064
58.	PREVAPP_DAYS_LAST_DUE_MEAN	0.0064
59.	PREVAPP_AMT_CREDIT_MAX	0.0064
60.	PREVAPP_DAYS_TERMINATION_MEAN	0.0063
61.	HOUR_APPR_PROCESS_START	0.0062
62.	PREVAPP_CNT_PAYMENT_MEAN	0.0061
63.	PREVAPP_AMT_DOWN_PAYMENT_MEAN	0.0060
64.	PREVAPP_AMT_APPLICATION_MAX	0.0060
65.	PREVAPP_RATE_DOWN_PAYMENT_MEAN	0.0059
66.	NAME_CONTRACT_STATUS_Active_x	0.0059
67.	PREVAPP_AMT_DOWN_PAYMENT_SUM	0.0058
68.	PREVAPP_DAYS_FIRST_DUE_MAX	0.0058
69.	PREVAPP_AMT_GOODS_PRICE_MAX	0.0057
70.	BUREAU_AMT_CREDIT_SUM_DEBT_MEAN	0.0057
71.	BUREAU_AMT_CREDIT_SUM_DEBT_SUM	0.0057
72.	NAME_CONTRACT_STATUS_Completed_x	0.0055
73.	BUREAU_DAYS_ENDDATE_FACT_MAX	0.0055
74.	PREVAPP_HOUR_APPR_PROCESS_START_MAX	0.0055
75.	BUREAU_AMT_CREDIT_SUM_DEBT_MAX	0.0054
76.	PREVAPP_RATE_DOWN_PAYMENT_MAX	0.0053

77. PREVAPP_AMT_DOWN_PAYMENT_MAX	0.0051
78. BUREAU_DAYS_ENDDATE_FACT_MIN	0.0050
79. TOTALAREA_MODE	0.0050
80. PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MAX	0.0049
81. LIVINGAREA_AVG	0.0048
82. YEARS_BEGINEXPLUATATION_AVG	0.0047
83. POS_MONTHS_BALANCE_MAX	0.0046
84. APARTMENTS_AVG	0.0045
85. CREDIT_ACTIVE_Closed	0.0045
86. CREDIT_ACTIVE_Active	0.0044
87. NAME_CONTRACT_STATUS_Refused_x	0.0044
88. NAME_CONTRACT_STATUS_Approved_x	0.0043
89. OWN_CAR_AGE	0.0043
90. CODE_REJECT_REASON_XAP	0.0041
91. NAME_YIELD_GROUP_high	0.0040
92. BUREAU_AMT_CREDIT_MAX_OVERDUE_MEAN	0.0039
93. LANDAREA_AVG	0.0039
94. OBS_60_CNT_SOCIAL_CIRCLE	0.0039
95. OBS_30_CNT_SOCIAL_CIRCLE	0.0039
96. POS_CNT_INSTALMENT_MAX	0.0037
97. BASEMENTAREA_AVG	0.0037
98. POS_CNT_INSTALMENT_FUTURE_MAX	0.0037
99. BUREAU_AMT_CREDIT_MAX_OVERDUE_MAX	0.0036
100. LIVINGAPARTMENTS_AVG	0.0036
101. BUREAU_AMT_CREDIT_MAX_OVERDUE_SUM	0.0036
102. CREDIT_TYPE_Credit card	0.0036
103. CREDIT_TYPE_Consumer credit	0.0036
104. POS_SK_DPD_DEF_MEAN	0.0036
105. YEARS_BUILD_AVG	0.0036
106. PREVAPP_DAYS_LAST_DUE_MAX	0.0035
107. NAME_PAYMENT_TYPE_XNA	0.0034
108. NAME_YIELD_GROUP_XNA	0.0034
109. NAME_PAYMENT_TYPE_Cash through the bank	0.0034
110. NAME_YIELD_GROUP_middle	0.0034
111. PREVAPP_DAYS_TERMINATION_MAX	0.0033
112. PREVAPP_CNT_PAYMENT_MAX	0.0033
113. PREVAPP_DAYS_FIRST_DRAWING_MEAN	0.0033
114. POS_SK_DPD_MEAN	0.0032
115. WEEKDAY_APPR_PROCESS_START_THURSDAY_y	0.0031
116. WEEKDAY_APPR_PROCESS_START_MONDAY_y	0.0031
117. AMT_REQ_CREDIT_BUREAU_YEAR	0.0031
118. NAME_CLIENT_TYPE_Repeater	0.0031
119. NONLIVINGAREA_AVG	0.0031
120. NAME_PORTFOLIO_POS	0.0031
121. NAME_PRODUCT_TYPE_walk-in	0.0031
122. WEEKDAY_APPR_PROCESS_START_WEDNESDAY_y	0.0031
123. CHANNEL_TYPE_Country-wide	0.0030
124. NAME_CONTRACT_TYPE_Cash loans	0.0030
125. CODE_REJECT_REASON_HC	0.0030
126. WEEKDAY_APPR_PROCESS_START_TUESDAY_y	0.0030
127. COMMONAREA_MODE	0.0030
128. WEEKDAY_APPR_PROCESS_START_FRIDAY	0.0030
129. ENTRANCES_AVG	0.0030
130. NAME_PRODUCT_TYPE_XNA	0.0030
131. NAME_CLIENT_TYPE_New	0.0030
132. PREVAPP_NFLAG_INSURED_ON_APPROVAL_MEAN	0.0030
133. CHANNEL_TYPE_Credit and cash offices	0.0030
134. COMMONAREA_AVG	0.0030
135. COMMONAREA_MEDI	0.0030
136. NAME_SELLER_INDUSTRY_Connectivity	0.0029
137. NAME_GOODS_CATEGORY_XNA	0.0029
138. NAME_SELLER_INDUSTRY_XNA	0.0029
139. NAME_YIELD_GROUP_low_normal	0.0029
140. NAME_SELLER_INDUSTRY_Consumer electronics	0.0029
141. NAME_PORTFOLIO_XNA	0.0029
142. CNT_FAM_MEMBERS	0.0029
143. NAME_CONTRACT_TYPE_Consumer loans	0.0029
144. NAME_PRODUCT_TYPE_x-sell	0.0029
145. NAME_CONTRACT_STATUS_Signed_x	0.0028
146. NAME_TYPE_SUITE_Unaccompanied_y	0.0028
147. WEEKDAY_APPR_PROCESS_START_SATURDAY_y	0.0028
148. NAME_PORTFOLIO_Cash	0.0028
149. POS_SK_DPD_SUM	0.0028
150. NAME_CONTRACT_TYPE_Revolving loans	0.0027
151. NAME_GOODS_CATEGORY_Mobile	0.0027
152. PRODUCT_COMBINATION_POS mobile with interest	0.0027
153. POS_SK_DPD_MAX	0.0026
154. POS_SK_DPD_DEF_SUM	0.0026

155.	POS_SK_DPD_DEF_MAX	0.0026
156.	PRODUCT_COMBINATION_Cash	0.0026
157.	NAME_CONTRACT_STATUS_Canceled_x	0.0026
158.	NAME_TYPE_SUITE_Family_y	0.0025
159.	NAME_PORTFOLIO_Cards	0.0025
160.	FLOORSMAX_AVG	0.0025
161.	CHANNEL_TYPE_Stone	0.0024
162.	WEEKDAY_APPR_PROCESS_START_SUNDAY_y	0.0024
163.	BUREAU_AMT_CREDIT_SUM_LIMIT_MEAN	0.0023
164.	BUREAU_AMT_CREDIT_SUM_LIMIT_SUM	0.0023
165.	CHANNEL_TYPE_AP+ (Cash loan)	0.0022
166.	NAME_GOODS_CATEGORY_Audio/Video	0.0022
167.	NAME_GOODS_CATEGORY_Computers	0.0022
168.	CODE_REJECT_REASON_LIMIT	0.0022
169.	CNT_CHILDREN	0.0021
170.	NAME_GOODS_CATEGORY_Consumer Electronics	0.0021
171.	DEF_30_CNT_SOCIAL_CIRCLE	0.0021
172.	BUREAU_AMT_CREDIT_SUM_LIMIT_MAX	0.0020
173.	FLOORSMIN_AVG	0.0020
174.	CODE_REJECT_REASON_SCOFR	0.0020
175.	CHANNEL_TYPE_Regional / Local	0.0020
176.	CHANNEL_TYPE_Contact center	0.0019
177.	NAME_CLIENT_TYPE_Refreshed	0.0019
178.	NAME_TYPE_SUITE_Spouse, partner_y	0.0018
179.	REGION_RATING_CLIENT	0.0018
180.	DEF_60_CNT_SOCIAL_CIRCLE	0.0018
181.	REGION_RATING_CLIENT_W_CITY	0.0017
182.	CREDIT_TYPE_Microloan	0.0017
183.	CODE_GENDER_M	0.0017
184.	PRODUCT_COMBINATION_Cash X-Sell: low	0.0016
185.	NONLIVINGAPARTMENTS_AVG	0.0016
186.	NAME_YIELD_GROUP_low_action	0.0015
187.	ELEVATORS_AVG	0.0014
188.	CODE_REJECT_REASON_SCO	0.0014
189.	NAME_EDUCATION_TYPE_Higher education	0.0014
190.	NAME_EDUCATION_TYPE_Secondary / secondary special	0.0014
191.	AMT_REQ_CREDIT_BUREAU_QRT	0.0014
192.	NAME_CASH_LOAN_PURPOSE_Repairs	0.0014
193.	NAME_INCOME_TYPE_Working	0.0013
194.	AMT_REQ_CREDIT_BUREAU_MON	0.0013
195.	NAME_GOODS_CATEGORY_Furniture	0.0013
196.	NAME_GOODS_CATEGORY_Photo / Cinema Equipment	0.0013
197.	NAME_TYPE_SUITE_Other_B_y	0.0013
198.	REG_CITY_NOT_WORK_CITY	0.0013
199.	NAME_CASH_LOAN_PURPOSE_Other	0.0012
200.	NAME_CONTRACT_STATUS_Returned to the store	0.0012

```
In [ ]: plt.figure(figsize=(12, 8))
top_features = feature_importance.head(364)

sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
plt.title('Top 20 Most Important Features (Random Forest)', fontsize=16, fontweight='bold')
plt.xlabel('Feature Importance', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.tight_layout()
plt.show()

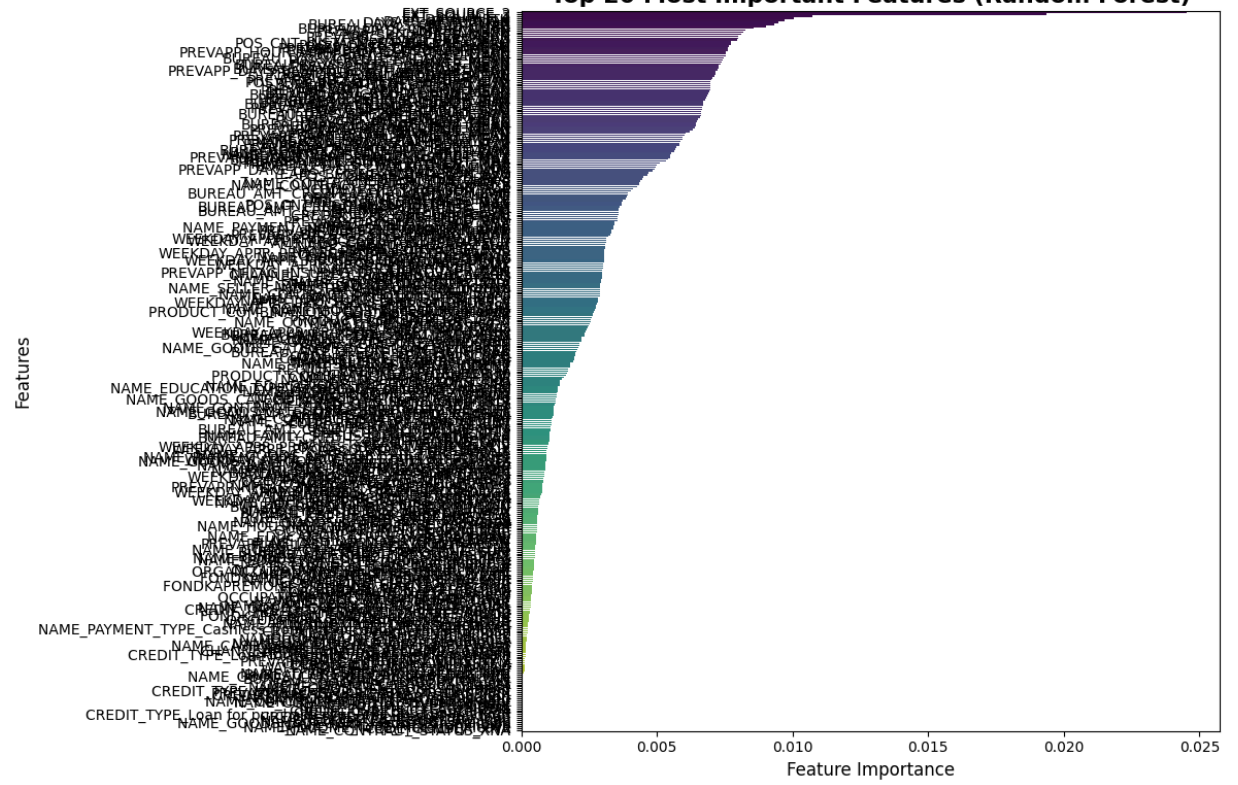
print("\nTop 550 Most Important Features:")
print("="*50)
for i, (idx, row) in enumerate(top_features.iterrows(), 1):
    print(f"{i:2d}. {row['feature']:<40} {row['importance']:.4f}")
```

<ipython-input-8-783812256>:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(data=top_features, y='feature', x='importance', palette='viridis')
```

Top 20 Most Important Features (Random Forest)



Top 550 Most Important Features:

=====		
1.	EXT_SOURCE_2	0.0245
2.	EXT_SOURCE_3	0.0193
3.	EXT_SOURCE_1	0.0107
4.	DAYS_BIRTH	0.0100
5.	DAYS_ID_PUBLISH	0.0097
6.	DAYS_REGISTRATION	0.0095
7.	BUREAU_DAYS_CREDIT_MAX	0.0093
8.	SK_ID_CURR	0.0090
9.	BUREAU_DAYS_CREDIT_MEAN	0.0086
10.	PREVAPP_SK_ID_PREV_MAX	0.0083
11.	AMT_ANNUITY	0.0082
12.	POS_SK_ID_PREV_MEAN	0.0081
13.	PREVAPP_SK_ID_PREV_MEAN	0.0080
14.	POS_SK_ID_PREV_MAX	0.0080
15.	DAYS_EMPLOYED	0.0080
16.	POS_CNT_INSTALMENT_FUTURE_MEAN	0.0077
17.	POS_MONTHS_BALANCE_SUM	0.0077
18.	DAYS_LAST_PHONE_CHANGE	0.0076
19.	PREVAPP_DAYS_DECISION_MEAN	0.0076
20.	PREVAPP_DAYS_DECISION_MAX	0.0076
21.	PREVAPP_HOUR_APPR_PROCESS_START_MEAN	0.0076
22.	BUREAU_DAYS_CREDIT_MIN	0.0075
23.	AGE	0.0075
24.	BUREAU_DAYS_CREDIT_ENDDATE_MEAN	0.0075
25.	POS_MONTHS_BALANCE_MEAN	0.0074
26.	AMT_CREDIT	0.0074
27.	BUREAU_DAYS_CREDIT_UPDATE_MEAN	0.0074
28.	PREVAPP_AMT_ANNUITY_MEAN	0.0073
29.	BUREAU_DAYS_CREDIT_ENDDATE_MAX	0.0073
30.	PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MEAN	0.0072
31.	PREVAPP_DAYS_FIRST_DUE_MEAN	0.0072
32.	REGION_POPULATION_RELATIVE	0.0072
33.	AMT_INCOME_TOTAL	0.0071
34.	BUREAU_SK_ID_BUREAU_MIN	0.0070
35.	PREVAPP_SELLERPLACE_AREA_MEAN	0.0070
36.	POS_CNT_INSTALMENT_FUTURE_SUM	0.0070
37.	BUREAU_AMT_CREDIT_SUM_SUM	0.0070
38.	PREVAPP_AMT_CREDIT_MEAN	0.0070
39.	BUREAU_SK_ID_BUREAU_MAX	0.0070
40.	PREVAPP_AMT_APPLICATION_MEAN	0.0070
41.	BUREAU_AMT_CREDIT_SUM_MEAN	0.0069
42.	BUREAU_DAYS_CREDIT_UPDATE_MIN	0.0069
43.	PREVAPP_AMT_ANNUITY_SUM	0.0068
44.	PREVAPP_SELLERPLACE_AREA_MAX	0.0068
45.	BUREAU_SK_ID_BUREAU_MEAN	0.0068
46.	PREVAPP_AMT_APPLICATION_SUM	0.0067
47.	BUREAU_DAYS_CREDIT_UPDATE_MAX	0.0067
48.	PREVAPP_AMT_GOODS_PRICE_SUM	0.0067
49.	PREVAPP_AMT_GOODS_PRICE_MEAN	0.0067
50.	POS_CNT_INSTALMENT_SUM	0.0067
51.	PREVAPP_AMT_CREDIT_SUM	0.0066
52.	BUREAU_DAYS_CREDIT_ENDDATE_MIN	0.0066
53.	BUREAU_AMT_CREDIT_SUM_MAX	0.0066
54.	AMT_GOODS_PRICE	0.0066
55.	POS_CNT_INSTALMENT_MEAN	0.0065
56.	PREVAPP_AMT_ANNUITY_MAX	0.0065
57.	BUREAU_DAYS_ENDDATE_FACT_MEAN	0.0064
58.	PREVAPP_DAYS_LAST_DUE_MEAN	0.0064
59.	PREVAPP_AMT_CREDIT_MAX	0.0064
60.	PREVAPP_DAYS_TERMINATION_MEAN	0.0063
61.	HOUR_APPR_PROCESS_START	0.0062
62.	PREVAPP_CNT_PAYMENT_MEAN	0.0061
63.	PREVAPP_AMT_DOWN_PAYMENT_MEAN	0.0060
64.	PREVAPP_AMT_APPLICATION_MAX	0.0060
65.	PREVAPP_RATE_DOWN_PAYMENT_MEAN	0.0059
66.	NAME_CONTRACT_STATUS_Active_x	0.0059
67.	PREVAPP_AMT_DOWN_PAYMENT_SUM	0.0058
68.	PREVAPP_DAYS_FIRST_DUE_MAX	0.0058
69.	PREVAPP_AMT_GOODS_PRICE_MAX	0.0057
70.	BUREAU_AMT_CREDIT_SUM_DEBT_MEAN	0.0057
71.	BUREAU_AMT_CREDIT_SUM_DEBT_SUM	0.0057
72.	NAME_CONTRACT_STATUS_Completed_x	0.0055
73.	BUREAU_DAYS_ENDDATE_FACT_MAX	0.0055
74.	PREVAPP_HOUR_APPR_PROCESS_START_MAX	0.0055
75.	BUREAU_AMT_CREDIT_SUM_DEBT_MAX	0.0054
76.	PREVAPP_RATE_DOWN_PAYMENT_MAX	0.0053

77. PREVAPP_AMT_DOWN_PAYMENT_MAX	0.0051
78. BUREAU_DAYS_ENDDATE_FACT_MIN	0.0050
79. TOTALAREA_MODE	0.0050
80. PREVAPP_DAYS_LAST_DUE_1ST_VERSION_MAX	0.0049
81. LIVINGAREA_AVG	0.0048
82. YEARS_BEGINEXPLUATATION_AVG	0.0047
83. POS_MONTHS_BALANCE_MAX	0.0046
84. APARTMENTS_AVG	0.0045
85. CREDIT_ACTIVE_Closed	0.0045
86. CREDIT_ACTIVE_Active	0.0044
87. NAME_CONTRACT_STATUS_Refused_x	0.0044
88. NAME_CONTRACT_STATUS_Approved_x	0.0043
89. OWN_CAR_AGE	0.0043
90. CODE_REJECT_REASON_XAP	0.0041
91. NAME_YIELD_GROUP_high	0.0040
92. BUREAU_AMT_CREDIT_MAX_OVERDUE_MEAN	0.0039
93. LANDAREA_AVG	0.0039
94. OBS_60_CNT_SOCIAL_CIRCLE	0.0039
95. OBS_30_CNT_SOCIAL_CIRCLE	0.0039
96. POS_CNT_INSTALMENT_MAX	0.0037
97. BASEMENTAREA_AVG	0.0037
98. POS_CNT_INSTALMENT_FUTURE_MAX	0.0037
99. BUREAU_AMT_CREDIT_MAX_OVERDUE_MAX	0.0036
100. LIVINGAPARTMENTS_AVG	0.0036
101. BUREAU_AMT_CREDIT_MAX_OVERDUE_SUM	0.0036
102. CREDIT_TYPE_Credit card	0.0036
103. CREDIT_TYPE_Consumer credit	0.0036
104. POS_SK_DPD_DEF_MEAN	0.0036
105. YEARS_BUILD_AVG	0.0036
106. PREVAPP_DAYS_LAST_DUE_MAX	0.0035
107. NAME_PAYMENT_TYPE_XNA	0.0034
108. NAME_YIELD_GROUP_XNA	0.0034
109. NAME_PAYMENT_TYPE_Cash through the bank	0.0034
110. NAME_YIELD_GROUP_middle	0.0034
111. PREVAPP_DAYS_TERMINATION_MAX	0.0033
112. PREVAPP_CNT_PAYMENT_MAX	0.0033
113. PREVAPP_DAYS_FIRST_DRAWING_MEAN	0.0033
114. POS_SK_DPD_MEAN	0.0032
115. WEEKDAY_APPR_PROCESS_START_THURSDAY_y	0.0031
116. WEEKDAY_APPR_PROCESS_START_MONDAY_y	0.0031
117. AMT_REQ_CREDIT_BUREAU_YEAR	0.0031
118. NAME_CLIENT_TYPE_Repeater	0.0031
119. NONLIVINGAREA_AVG	0.0031
120. NAME_PORTFOLIO_POS	0.0031
121. NAME_PRODUCT_TYPE_walk-in	0.0031
122. WEEKDAY_APPR_PROCESS_START_WEDNESDAY_y	0.0031
123. CHANNEL_TYPE_Country-wide	0.0030
124. NAME_CONTRACT_TYPE_Cash loans	0.0030
125. CODE_REJECT_REASON_HC	0.0030
126. WEEKDAY_APPR_PROCESS_START_TUESDAY_y	0.0030
127. COMMONAREA_MODE	0.0030
128. WEEKDAY_APPR_PROCESS_START_FRIDAY	0.0030
129. ENTRANCES_AVG	0.0030
130. NAME_PRODUCT_TYPE_XNA	0.0030
131. NAME_CLIENT_TYPE_New	0.0030
132. PREVAPP_NFLAG_INSURED_ON_APPROVAL_MEAN	0.0030
133. CHANNEL_TYPE_Credit and cash offices	0.0030
134. COMMONAREA_AVG	0.0030
135. COMMONAREA_MEDI	0.0030
136. NAME_SELLER_INDUSTRY_Connectivity	0.0029
137. NAME_GOODS_CATEGORY_XNA	0.0029
138. NAME_SELLER_INDUSTRY_XNA	0.0029
139. NAME_YIELD_GROUP_low_normal	0.0029
140. NAME_SELLER_INDUSTRY_Consumer electronics	0.0029
141. NAME_PORTFOLIO_XNA	0.0029
142. CNT_FAM_MEMBERS	0.0029
143. NAME_CONTRACT_TYPE_Consumer loans	0.0029
144. NAME_PRODUCT_TYPE_x-sell	0.0029
145. NAME_CONTRACT_STATUS_Signed_x	0.0028
146. NAME_TYPE_SUITE_Unaccompanied_y	0.0028
147. WEEKDAY_APPR_PROCESS_START_SATURDAY_y	0.0028
148. NAME_PORTFOLIO_Cash	0.0028
149. POS_SK_DPD_SUM	0.0028
150. NAME_CONTRACT_TYPE_Revolving loans	0.0027
151. NAME_GOODS_CATEGORY_Mobile	0.0027
152. PRODUCT_COMBINATION_POS mobile with interest	0.0027
153. POS_SK_DPD_MAX	0.0026
154. POS_SK_DPD_DEF_SUM	0.0026

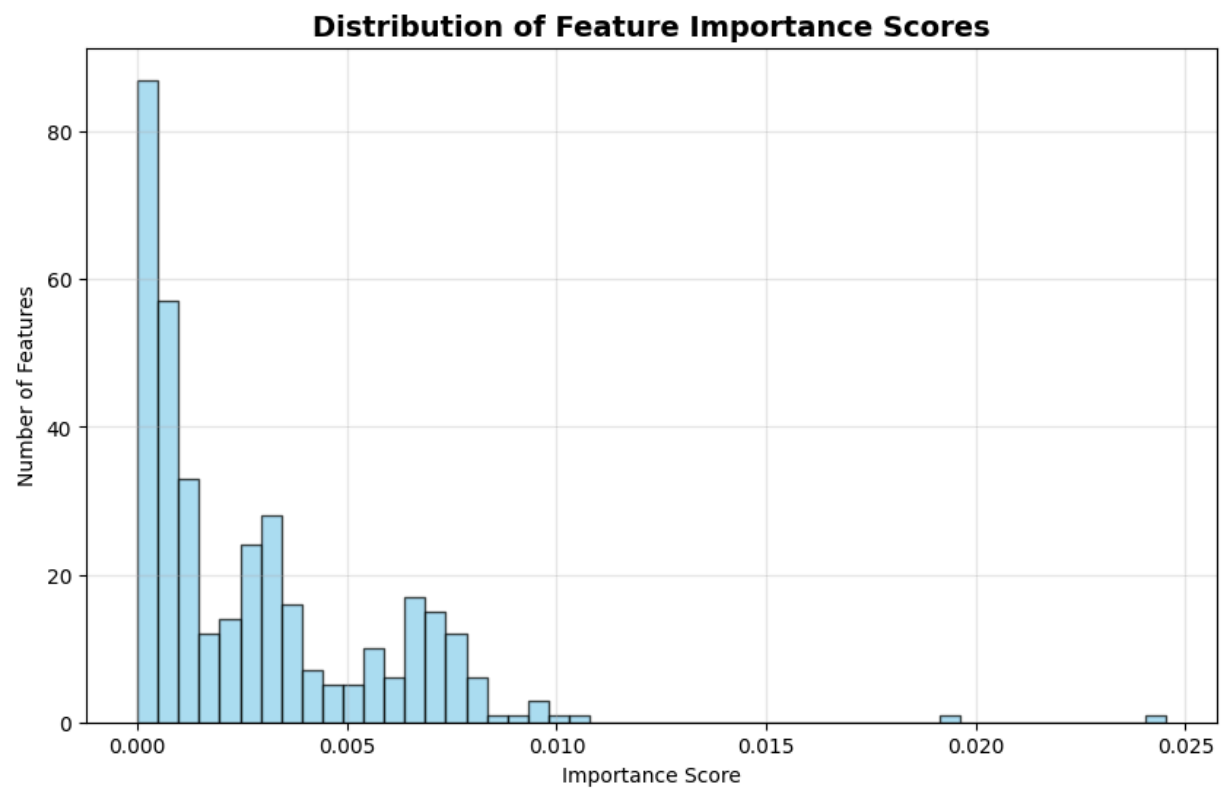
155.	POS_SK_DPD_DEF_MAX	0.0026
156.	PRODUCT_COMBINATION_Cash	0.0026
157.	NAME_CONTRACT_STATUS_Canceled_x	0.0026
158.	NAME_TYPE_SUITE_Family_y	0.0025
159.	NAME_PORTFOLIO_Cards	0.0025
160.	FLOORSMAX_AVG	0.0025
161.	CHANNEL_TYPE_Stone	0.0024
162.	WEEKDAY_APPR_PROCESS_START_SUNDAY_y	0.0024
163.	BUREAU_AMT_CREDIT_SUM_LIMIT_MEAN	0.0023
164.	BUREAU_AMT_CREDIT_SUM_LIMIT_SUM	0.0023
165.	CHANNEL_TYPE_AP+ (Cash loan)	0.0022
166.	NAME_GOODS_CATEGORY_Audio/Video	0.0022
167.	NAME_GOODS_CATEGORY_Computers	0.0022
168.	CODE_REJECT_REASON_LIMIT	0.0022
169.	CNT_CHILDREN	0.0021
170.	NAME_GOODS_CATEGORY_Consumer Electronics	0.0021
171.	DEF_30_CNT_SOCIAL_CIRCLE	0.0021
172.	BUREAU_AMT_CREDIT_SUM_LIMIT_MAX	0.0020
173.	FLOORSMIN_AVG	0.0020
174.	CODE_REJECT_REASON_SCOFR	0.0020
175.	CHANNEL_TYPE_Regional / Local	0.0020
176.	CHANNEL_TYPE_Contact center	0.0019
177.	NAME_CLIENT_TYPE_Refreshed	0.0019
178.	NAME_TYPE_SUITE_Spouse, partner_y	0.0018
179.	REGION_RATING_CLIENT	0.0018
180.	DEF_60_CNT_SOCIAL_CIRCLE	0.0018
181.	REGION_RATING_CLIENT_W_CITY	0.0017
182.	CREDIT_TYPE_Microloan	0.0017
183.	CODE_GENDER_M	0.0017
184.	PRODUCT_COMBINATION_Cash X-Sell: low	0.0016
185.	NONLIVINGAPARTMENTS_AVG	0.0016
186.	NAME_YIELD_GROUP_low_action	0.0015
187.	ELEVATORS_AVG	0.0014
188.	CODE_REJECT_REASON_SCO	0.0014
189.	NAME_EDUCATION_TYPE_Higher education	0.0014
190.	NAME_EDUCATION_TYPE_Secondary / secondary special	0.0014
191.	AMT_REQ_CREDIT_BUREAU_QRT	0.0014
192.	NAME_CASH_LOAN_PURPOSE_Repairs	0.0014
193.	NAME_INCOME_TYPE_Working	0.0013
194.	AMT_REQ_CREDIT_BUREAU_MON	0.0013
195.	NAME_GOODS_CATEGORY_Furniture	0.0013
196.	NAME_GOODS_CATEGORY_Photo / Cinema Equipment	0.0013
197.	NAME_TYPE_SUITE_Other_B_y	0.0013
198.	REG_CITY_NOT_WORK_CITY	0.0013
199.	NAME_CASH_LOAN_PURPOSE_Other	0.0012
200.	NAME_CONTRACT_STATUS_Returned to the store	0.0012
201.	REG_CITY_NOT_LIVE_CITY	0.0012
202.	NAME_GOODS_CATEGORY_Construction Materials	0.0012
203.	BUREAU_AMT_CREDIT_SUM_OVERDUE_MEAN	0.0012
204.	NAME_TYPE_SUITE_Children	0.0012
205.	NAME_FAMILY_STATUS_Married	0.0011
206.	NAME_CONTRACT_STATUS_Unused offer	0.0011
207.	NAME_SELLER_INDUSTRY_Construction	0.0011
208.	CODE_REJECT_REASON_CLIENT	0.0011
209.	FLAG_OWN_REALTY	0.0011
210.	OCCUPATION_TYPE_Laborers	0.0011
211.	BUREAU_AMT_CREDIT_SUM_OVERDUE_SUM	0.0011
212.	LIVE_CITY_NOT_WORK_CITY	0.0011
213.	FLAG_DOCUMENT_3	0.0011
214.	NAME_FAMILY_STATUS_Single / not married	0.0011
215.	BUREAU_AMT_CREDIT_SUM_OVERDUE_MAX	0.0010
216.	FLAG_OWN_CAR	0.0010
217.	CREDIT_ACTIVE_Sold	0.0010
218.	FLAG_WORK_PHONE	0.0010
219.	NAME_TYPE_SUITE_Other_A_y	0.0010
220.	WEEKDAY_APPR_PROCESS_START_WEDNESDAY_x	0.0010
221.	WEEKDAY_APPR_PROCESS_START_THURSDAY_x	0.0010
222.	OCCUPATION_TYPE_Drivers	0.0009
223.	FLAG_PHONE	0.0009
224.	NAME_GOODS_CATEGORY_Auto Accessories	0.0009
225.	NAME_PAYMENT_TYPE_Non-cash from your account	0.0009
226.	WEEKDAY_APPR_PROCESS_START_TUESDAY_x	0.0009
227.	NAME_GOODS_CATEGORY_Clothing and Accessories	0.0009
228.	CREDIT_TYPE_Car loan	0.0009
229.	NAME_INCOME_TYPE_Commercial associate	0.0009
230.	NAME_TYPE_SUITE_Unaccompanied_x	0.0009
231.	OCCUPATION_TYPE_Sales staff	0.0009
232.	NAME_HOUSING_TYPE_House / apartment	0.0009

233.	BUREAU_CNT_CREDIT_PROLONG_MEAN	0.0009
234.	NAME_HOUSING_TYPE_With parents	0.0009
235.	WEEKDAY_APPR_PROCESS_START_MONDAY_x	0.0008
236.	NAME_GOODS_CATEGORY_Jewelry	0.0008
237.	OCCUPATION_TYPE_Low-skill Laborers	0.0008
238.	WALLSMATERIAL_MODE_Panel	0.0008
239.	NAME_CONTRACT_STATUS_Approved_y	0.0008
240.	PREVAPP_NFLAG_INSURED_ON_APPROVAL_MAX	0.0008
241.	NAME_TYPE_SUITE_Family_x	0.0008
242.	NAME_GOODS_CATEGORY_Vehicles	0.0008
243.	WEEKDAY_APPR_PROCESS_START_SATURDAY_x	0.0008
244.	NAME_FAMILY_STATUS_Separated	0.0008
245.	WALLSMATERIAL_MODE_Stone, brick	0.0007
246.	CODE_REJECT_REASON_XNA	0.0007
247.	WEEKDAY_APPR_PROCESS_START_SUNDAY_x	0.0007
248.	NAME_HOUSING_TYPE_Rented apartment	0.0007
249.	OCCUPATION_TYPE_Security staff	0.0007
250.	BUREAU_CREDIT_DAY_OVERDUE_MEAN	0.0006
251.	REG_REGION_NOT_WORK_REGION	0.0006
252.	NAME_TYPE_SUITE_Spouse, partner_x	0.0006
253.	BUREAU_CREDIT_DAY_OVERDUE_MAX	0.0006
254.	AMT_REQ_CREDIT_BUREAU_WEEK	0.0006
255.	FLAG_DOCUMENT_8	0.0006
256.	BUREAU_CREDIT_DAY_OVERDUE_SUM	0.0006
257.	CREDIT_TYPE_Mortgage	0.0006
258.	NAME_GOODS_CATEGORY_Homewares	0.0006
259.	NAME_INCOME_TYPE_Pensioner	0.0006
260.	NAME_HOUSING_TYPE_Municipal apartment	0.0006
261.	OCCUPATION_TYPE_Core staff	0.0006
262.	OCCUPATION_TYPE_Cooking staff	0.0006
263.	FLAG_EMP_PHONE	0.0006
264.	OCCUPATION_TYPE_Managers	0.0006
265.	NAME_EDUCATION_TYPE_Lower secondary	0.0006
266.	FLAG_EMAIL	0.0006
267.	LIVE_REGION_NOT_WORK_REGION	0.0005
268.	FLAG_LAST_APPL_PER_CONTRACT_Y	0.0005
269.	PREVAPP_NFLAG_LAST_APPL_IN_DAY_MEAN	0.0005
270.	FLAG_LAST_APPL_PER_CONTRACT_N	0.0005
271.	NAME_CONTRACT_TYPE	0.0005
272.	NAME_GOODS_CATEGORY_Sport and Leisure	0.0005
273.	BUREAU_CNT_CREDIT_PROLONG_SUM	0.0005
274.	ORGANIZATION_TYPE_Government	0.0005
275.	BUREAU_CNT_CREDIT_PROLONG_MAX	0.0005
276.	NAME_GOODS_CATEGORY_Office Appliances	0.0005
277.	NAME_EDUCATION_TYPE_Incomplete higher	0.0005
278.	NAME_TYPE_SUITE_Group of people_y	0.0005
279.	FLAG_DOCUMENT_6	0.0005
280.	OCCUPATION_TYPE_Cleaning staff	0.0005
281.	NAME_FAMILY_STATUS_Widow	0.0005
282.	OCCUPATION_TYPE_High skill tech staff	0.0005
283.	ORGANIZATION_TYPE_Business Entity Type 1	0.0004
284.	NAME_INCOME_TYPE_State servant	0.0004
285.	CODE_REJECT_REASON_VERIF	0.0004
286.	FONDKAPREMONT_MODE_reg oper account	0.0004
287.	NAME_GOODS_CATEGORY_Gardening	0.0004
288.	OCCUPATION_TYPE_Medicine staff	0.0004
289.	NAME_CLIENT_TYPE_XNA	0.0004
290.	FONDKAPREMONT_MODE_reg oper spec account	0.0004
291.	REG_REGION_NOT_LIVE_REGION	0.0004
292.	ORGANIZATION_TYPE_School	0.0004
293.	ORGANIZATION_TYPE_Medicine	0.0004
294.	WALLSMATERIAL_MODE_Wooden	0.0004
295.	NAME_TYPE_SUITE_Other_B_x	0.0004
296.	OCCUPATION_TYPE_Waiters/barmen staff	0.0004
297.	NAME_GOODS_CATEGORY_Other	0.0004
298.	NAME_GOODS_CATEGORY_Medicine	0.0004
299.	AMT_REQ_CREDIT_BUREAU_DAY	0.0004
300.	NAME_CASH_LOAN_PURPOSE_Education	0.0004
301.	NAME_GOODS_CATEGORY_Medical Supplies	0.0004
302.	CREDIT_TYPE_Loan for business development	0.0003
303.	HOUSETYPE_MODE_specific housing	0.0003
304.	EMERGENCYSTATE_MODE	0.0003
305.	FONDKAPREMONT_MODE_org spec account	0.0003
306.	WALLSMATERIAL_MODE_Others	0.0003
307.	OCCUPATION_TYPE_Private service staff	0.0003
308.	AMT_REQ_CREDIT_BUREAU_HOUR	0.0003
309.	NAME_HOUSING_TYPE_Office apartment	0.0003
310.	OCCUPATION_TYPE_Secretaries	0.0003

311. WALLSMATERIAL_MODE_Mixed	0.0003
312. NAME_PAYMENT_TYPE_Cashless from the account of the employer	0.0002
313. CREDIT_TYPE_Another type of loan	0.0002
314. FLAG_CONT_MOBILE	0.0002
315. NAME_TYPE_SUITE_Other_A_x	0.0002
316. OCCUPATION_TYPE_Realty agents	0.0002
317. NAME_CONTRACT_STATUS_Demand_x	0.0002
318. HOUSETYPE_MODE_terraced house	0.0002
319. NAME_GOODS_CATEGORY_Direct Sales	0.0002
320. NAME_CASH_LOAN_PURPOSE_Buying a new car	0.0002
321. ORGANIZATION_TYPE_Trade: type 1	0.0002
322. NAME_GOODS_CATEGORY_Tourism	0.0002
323. CHANNEL_TYPE_Channel of corporate sales	0.0002
324. CREDIT_TYPE_Unknown type of loan	0.0002
325. CREDIT_TYPE_Loan for working capital replenishment	0.0002
326. CREDIT_CURRENCY_currency 1	0.0002
327. CREDIT_CURRENCY_currency 2	0.0001
328. PREVAPP_DAYS_FIRST_DRAWING_MAX	0.0001
329. CODE_REJECT_REASON_SYSTEM	0.0001
330. CREDIT_CURRENCY_currency 3	0.0001
331. WALLSMATERIAL_MODE_Monolithic	0.0001
332. OCCUPATION_TYPE_IT staff	0.0001
333. OCCUPATION_TYPE_HR staff	0.0001
334. NAME_TYPE_SUITE_Group of people_x	0.0001
335. BUREAU_CREDIT_DAY_OVERDUE_MIN	0.0001
336. NAME_GOODS_CATEGORY_Additional Service	0.0001
337. BUREAU_CNT_CREDIT_PROLONG_MIN	0.0001
338. NAME_PORTFOLIO_Cars	0.0001
339. NAME_GOODS_CATEGORY_Fitness	0.0000
340. CHANNEL_TYPE_Car dealer	0.0000
341. NAME_GOODS_CATEGORY_Insurance	0.0000
342. NAME_GOODS_CATEGORY_Weapon	0.0000
343. CREDIT_TYPE_Loan for the purchase of equipment	0.0000
344. PREVAPP_NFLAG_LAST_APPL_IN_DAY_MAX	0.0000
345. CREDIT_TYPE_Cash loan (non-earmarked)	0.0000
346. NAME_GOODS_CATEGORY_Education	0.0000
347. NAME_INCOME_TYPE_Unemployed	0.0000
348. NAME_CONTRACT_STATUS_Amortized debt	0.0000
349. NAME_CONTRACT_STATUS_Canceled_y	0.0000
350. CREDIT_ACTIVE_Bad debt	0.0000
351. NAME_INCOME_TYPE_Student	0.0000
352. FLAG_MOBIL	0.0000
353. CREDIT_CURRENCY_currency 4	0.0000
354. CREDIT_TYPE_Interbank credit	0.0000
355. CREDIT_TYPE_Loan for purchase of shares (margin lending)	0.0000
356. CREDIT_TYPE_Mobile operator loan	0.0000
357. CREDIT_TYPE_Real estate loan	0.0000
358. NAME_GOODS_CATEGORY_Animals	0.0000
359. NAME_GOODS_CATEGORY_House Construction	0.0000
360. NAME_FAMILY_STATUS_Unknown	0.0000
361. NAME_INCOME_TYPE_Maternity leave	0.0000
362. CODE_GENDER_XNA	0.0000
363. NAME_CONTRACT_STATUS_XNA	0.0000

```
In [ ]: # Feature importance distribution
plt.figure(figsize=(10, 6))
plt.hist(feature_importance['importance'], bins=50, alpha=0.7, color='skyblue', edgecolor='black')
plt.title('Distribution of Feature Importance Scores', fontsize=14, fontweight='bold')
plt.xlabel('Importance Score')
plt.ylabel('Number of Features')
plt.grid(True, alpha=0.3)
plt.show()

# Show cumulative importance
feature_importance['cumulative_importance'] = feature_importance['importance'].cumsum()
print(f"\nTop 10 features account for {feature_importance.head(10)['cumulative_importance'].iloc[-1]:.3f} of")
print(f"Top 20 features account for {feature_importance.head(20)['cumulative_importance'].iloc[-1]:.3f} of to")
print(f"Top 50 features account for {feature_importance.head(50)['cumulative_importance'].iloc[-1]:.3f} of to")
```



Top 10 features account for 0.119 of total importance
Top 20 features account for 0.198 of total importance
Top 50 features account for 0.410 of total importance

```
In [ ]: # Show cumulative importance
feature_importance['cumulative_importance'] = feature_importance['importance'].cumsum()
print(f"\nTop 100 features account for {feature_importance.head(100)['cumulative_importance'].iloc[-1]:.3f} of total importance")
print(f"Top 175 features account for {feature_importance.head(175)['cumulative_importance'].iloc[-1]:.3f} of total importance")
print(f"Top 300 features account for {feature_importance.head(300)['cumulative_importance'].iloc[-1]:.3f} of total importance")
```

Top 100 features account for 0.670 of total importance
Top 175 features account for 0.883 of total importance
Top 300 features account for 0.992 of total importance